



Override and underperform

The empirical cost of human discretion in rule-based trading systems

We synthesise the empirical literature on the performance gap between rule-based trading systems run mechanically and the same systems run with discretionary override permitted to the trader. Across studies covering managed-futures CTAs, prop-trading desks, retail systematic traders, and the original Turtle program, the consistent finding is that human override of mechanical rules costs approximately 200–400 basis points of annualised return, with the gap widening during stress periods when override feels most justified. The mechanism appears to be a combination of overconfidence bias (Odean, 1998), recency-driven recalibration (Tversky and Kahneman, 1974), loss aversion expressed as premature exit (Shefrin and Statman, 1985), and confirmation bias in the selective interpretation of recent signal performance (Klayman and Ha, 1987). The underperformance is not driven by single catastrophic overrides but by a pattern of small, individually-justifiable adjustments that cumulatively produce systematic drag. We document the cross-study consistency of the finding, examine the conditions under which override has positive expected value (rare and specifically characterised), and discuss the design implications for systems that aim to capture documented backtested returns in live trading.

KEY FINDINGS

I. Override consistently costs 200–400bp per year. Across studies of CTAs, prop traders, retail systematic traders, and the Turtle program, mechanical execution outperforms discretionary override by 200–400bp annualised.

II. The gap widens precisely when override feels most justified. The largest underperformance from override occurs during stress periods (drawdowns, regime shifts, news shocks) — exactly when the trader's confidence in their override judgement is highest.

III. Damage is cumulative, not catastrophic. Override underperformance is not driven by single bad

decisions but by hundreds of small, individually-defensible adjustments that produce systematic drag.

IV. The behavioural mechanisms are well-documented. Overconfidence (Odean), recency (Tversky-Kahneman), loss aversion (Shefrin-Statman), and confirmation bias (Klayman-Ha) all contribute. None is unique to trading; all are reliably present.

V. The system-design implication is to remove the option. If override produces predictable underperformance, the most reliable way to capture documented system returns is to make override unavailable in the live implementation — not to discipline the trader's discretion.

EXEC. Executive summary

A systematic trading rule run mechanically and the same rule run with discretionary override permitted are not the same strategy. Across multiple studies covering different trader populations and time periods, the discretionary version reliably underperforms the mechanical version by 200–400 basis points annualised — a gap larger than the equity premium itself.

This note synthesises the empirical literature on the cost of human override in rule-based trading systems. The relevant studies span four trader populations: managed-futures CTAs (where most strategies are systematic but discretionary override is widely available), prop-trading desks (where individual traders frequently override systematic signals), retail systematic-strategy users (where the trader chooses whether to take each signal), and the documented case of the original Turtle program (where two cohorts received identical rules but produced dramatically different live performance). Across all four populations, the same pattern emerges: traders who override systematic signals underperform the mechanical version of those signals by a consistent and economically substantial margin.

The headline magnitude — 200–400bp per year — is striking. The U.S. equity premium over the long run is approximately 5%; the override cost erases more than half of it. A retail trader executing a strategy with documented 8% Sharpe-1 historical performance who overrides 25% of signals is, in expectation, capturing approximately 5% live performance — meaningfully worse than passive index investment.

The mechanism is well-understood in behavioural finance. [Odean \(1998\)](#) documented systematic overconfidence in retail trading decisions, with self-attribution of past success leading to over-trading and over-confidence in subsequent discretionary judgements. [Tversky and Kahneman \(1974\)](#) established that human judgement weights recent information more heavily than warranted, causing overrides to be driven by recent strategy losses (just as the strategy is most likely to mean-revert). [Shefrin and Statman \(1985\)](#) documented the disposition effect — premature exit from winners, prolonged holding of losers — which manifests in override behaviour as cutting profitable systematic positions short. [Klayman and Ha \(1987\)](#) documented confirmation bias in the selective evaluation of recent signal performance.

The most striking feature of the override-cost literature is the consistency of the finding across study designs. The cost is not driven by a few catastrophic decisions; it is driven by hundreds of small, individually-defensible adjustments that cumulatively produce systematic drag. Each individual override appears reasonable in the moment; the aggregate distribution of overrides exhibits the predictable behavioural biases listed above.

The system-design implication is direct. If override produces predictable underperformance, the most reliable way to capture documented backtested returns in live implementation is to make override unavailable. Asking the trader to be disciplined enough to refrain from override in the moments when it is most tempting is asking for behavioural performance that the literature suggests is rare. Mechanical execution — the same code, running on the same broker, without manual intervention — is the only reliable mechanism for closing the gap between backtest and live performance.

01. Introduction

The intuitive appeal of discretionary override is straightforward. A systematic rule is a static specification, optimised on historical data; markets evolve and circumstances change, and a competent trader observing the live tape should be able to identify situations in which the rule's prescription is sub-optimal. Override permits the trader to capture this judgemental value-add while retaining the systematic framework as the default.

The empirical record contradicts this intuition. Studies of trader populations who have access to discretionary override consistently find that the override option is exercised in a manner that destroys value rather than creating it. The pattern is robust across populations, time periods, asset classes, and strategy types. The override cost, measured in basis points of annualised return, ranges from 200 to 400 across studies.

The literature on this topic is large but somewhat fragmented across disciplines. The behavioural finance literature has documented the underlying biases (overconfidence, recency, loss aversion, confirmation bias) without specifically connecting them to trading-system override. The trading practitioner literature has documented the empirical cost of override without systematically connecting it to the underlying behavioural mechanisms. The academic finance literature has touched on the topic primarily through studies of mutual fund manager skill versus luck ([Fama and French, 2010](#)) and active-versus-passive performance comparisons.

This note synthesises the relevant evidence into a coherent treatment. We focus on three questions: (1) What is the empirical magnitude of the override cost across studies? (2) What behavioural mechanisms drive the cost? (3) What system-design implications follow from the empirical magnitude and the mechanisms?

The note proceeds as follows. Section 2 reviews the major studies documenting override cost. Section 3 examines the behavioural mechanisms. Section 4 discusses the conditions under which override has positive expected value (rare, but identifiable). Section 5 covers system-design implications.

02. The empirical record

2.1 The Turtle program

The clearest natural experiment in trading-system override is the Turtle program of Richard Dennis and William Eckhardt in 1983. Two cohorts of approximately fifteen traders each received identical training in a specific systematic trend-following methodology. The mechanical rules were the same across all traders; the only variable was each trader's individual willingness to follow them in live trading.

The performance dispersion across the cohort over the subsequent four years was substantial. The most successful traders (Jerry Parker, Curt Faith) compounded at 30%+ annualised over multi-year periods; the least successful produced returns close to zero or modestly negative. [Faith \(2007\)](#), who was the most successful trainee, attributes the dispersion almost entirely to override behaviour: the traders who deviated from the rules produced substantially worse outcomes than those who did not. The mechanical version of the rules, applied without override, would have produced the upper-quartile performance for every trainee in the cohort.

The Turtle case is particularly clean because the rules were specified, the training was uniform, and the trader population was selected for systematic discipline. The performance gap therefore cannot be attributed to differential rule quality, training quality, or trader selection. It must be attributed to the differential exercise of override. The estimated cost of override in the Turtle cohort is approximately 400bp annualised between the disciplined and undisciplined trader subsets.

2.2 Managed-futures CTA studies

[Fung and Hsieh \(2001\)](#) document the performance characteristics of CTAs running systematic strategies, with particular attention to the relationship between rule discipline and realised performance. Their cross-sectional

analysis suggests that CTAs reporting more rigorous adherence to systematic rules produce higher Sharpe ratios and lower drawdowns than CTAs reporting more discretionary deviation, controlling for strategy type and asset class. The implied override cost in their analysis is approximately 250bp annualised.

[Bhansali and Davis \(2010\)](#) extend the analysis to a longer time period (1994–2008) and a broader CTA universe. They find that the override cost is concentrated in stress periods: during normal market conditions, the gap between mechanical-discipline and discretionary-override CTAs is small (50–100bp); during stress periods (1998 LTCM, 2002 dot-com aftermath, 2008 GFC), the gap widens to 600–800bp. The asymmetry is consistent with the behavioural mechanism: override is most damaging precisely when the trader's confidence in their judgement is highest.

2.3 Prop-desk studies

[Coval and Shumway \(2005\)](#) studied a sample of prop-desk traders at a major U.S. financial institution, with detailed transaction data permitting analysis of the relationship between trader-initiated overrides and performance. Their finding: traders who deviated more frequently from systematic signals produced significantly lower realised Sharpe ratios than traders who followed signals more mechanically, controlling for tenure, asset class, and overall trading volume.

The Coval-Shumway study is methodologically clean because it uses individual-trader data with a clear definition of "override" (deviation from a desk-level systematic signal). The estimated override cost in their sample is approximately 220bp annualised, with the gap widening to 350bp for traders in the bottom quartile of risk-adjusted performance.

2.4 Retail systematic trader studies

[Barber and Odean \(2000\)](#) analyse the trading records of approximately 66,000 retail brokerage accounts at a major U.S. discount broker over the 1991–1996 period. The finding most relevant to override-cost analysis: the most-active traders (those most willing to deviate from any disciplined approach) underperform the most-passive traders by approximately 600bp annualised. The performance gap is driven primarily by transaction costs at the highest activity level, but the residual after accounting for transaction costs is approximately 200bp — consistent with the override-cost magnitude observed in other populations.

More recent studies of retail systematic-strategy users (cited in [Bhansali-Davis, appendix](#)) find similar patterns: users who execute systematic strategies with override permission underperform mechanical execution by 250–400bp annualised over multi-year periods.

03. Behavioural mechanisms

The empirical consistency of override cost across populations and time periods suggests a structural rather than coincidental cause. Behavioural finance has identified four primary mechanisms, each well-documented in non-trading contexts and demonstrably present in trading override decisions.

3.1 Overconfidence

[Odean \(1998\)](#) and [Barber and Odean \(2001\)](#) document that retail traders systematically over-estimate the precision of their own information and judgement. The mechanism is amplified by self-attribution: profitable trades are attributed to skill, unprofitable trades to bad luck or external factors, producing a one-way ratchet in confidence over time. [Glaser and Weber \(2007\)](#) extend this to professional traders, finding similar patterns in survey-based and behavioural measures.

In the override context, overconfidence manifests as an overestimate of the value-added of the trader's discretionary judgement relative to the systematic signal. The trader believes their override is informed by their reading of "the situation"; the empirical record suggests their reading of "the situation" is itself biased and adds noise rather than signal.

3.2 Recency bias

[Tversky and Kahneman \(1974\)](#) documented that human judgement weights recent information more heavily than warranted by its statistical relevance. [De Bondt and Thaler \(1985\)](#) applied this to financial markets, showing that investors over-react to short-term performance information and under-react to longer-term mean-reversion patterns.

In the override context, recency bias drives the systematic pattern in which traders increase override frequency after recent strategy losses (just as the strategy is most likely to mean-revert favourably) and decrease override after recent gains (just as the strategy is most likely to face an adverse run). The pattern produces an "anti-momentum" exposure in the override decisions that destroys value relative to the underlying systematic strategy.

3.3 Loss aversion and the disposition effect

[Shefrin and Statman \(1985\)](#) document that investors are reluctant to realise losses and quick to realise gains — the disposition effect. [Grinblatt and Han \(2005\)](#) extend this to the cross-section of investor behaviour and show its impact on momentum effects in equity markets.

In the override context, loss aversion manifests as premature exit from systematic positions that have moved against the trader. The trader, observing a losing position, overrides the system's "hold" or "add" prescription with a "close" decision, locking in the loss before the system's expected recovery has time to materialise. The aggregate effect is to systematically curtail the right tail of the strategy's return distribution while preserving the left tail.

3.4 Confirmation bias

[Klayman and Ha \(1987\)](#) document the systematic tendency for humans to seek and weight evidence that confirms prior beliefs while underweighting evidence that contradicts them. In trading, this manifests as selective interpretation of recent signal performance: an underperforming systematic rule is "confirmed" as broken by the next adverse outcome, while contradicting outcomes are dismissed as noise.

The combined effect of these four mechanisms produces a predictable pattern of override behaviour that systematically destroys rather than creates value. The pattern is not unique to any individual trader or trader type; it is, to a first approximation, a feature of human cognition that operates regardless of the trader's intelligence, training, or discipline.

04. When override has positive expected value

The empirical record on override cost is sufficiently strong that the natural question is whether there are conditions under which override does add value. The answer is "rarely, and specifically." Three conditions roughly characterise the cases:

Operational anomalies. When the systematic rule's input data is corrupted (bad price feed, missing data, gap from a corporate action), discretionary override to halt or adjust the trade is value-creating. These cases are typically detectable through validation rules rather than requiring trader judgement.

Discrete regime breaks. When a major structural event (currency revaluation, central bank intervention, major regulatory change) materially alters the market conditions under which the rule was developed, halting the rule pending review may be value-creating. Distinguishing such events from ordinary news shocks is, however, difficult ex ante; most "regime breaks" identified in real-time turn out to be noise.

Information advantage. When the trader has access to information not reflected in the systematic signal (proprietary fundamental research, private order flow, insider knowledge — to the extent legal), override may add value. The condition is restrictive: most trader perceptions of "information advantage" are illusory, with the perceived information already reflected in the price.

The cumulative occurrence rate of these legitimate-override conditions is estimated at less than 5% of trade decisions in most studies. The remaining 95%+ of override decisions are driven by the behavioural mechanisms discussed above and produce systematic value destruction.

05. Design implications

5.1 *The mechanical-execution principle*

If override has 5% legitimate occurrence and 95% behaviourally-driven occurrence, and if the average behavioural override destroys value while the average legitimate override preserves it, the expected value of permitting override is overwhelmingly negative. The system-design implication is direct: do not permit override.

The implementation of this principle varies across institutional contexts. For an institutional CTA, mechanical execution is achieved through risk-management infrastructure that flags and pre-approves any deviation from the systematic signal. For a prop desk, it is achieved through compliance and management oversight. For a retail systematic-strategy user, it is most reliably achieved by automating the strategy entirely — having the broker's API receive signals directly without manual intervention from the trader.

The retail case is the most challenging. Most retail brokerages permit manual order entry alongside automated signals; the trader retains the option to override even within an "automated" framework. The most reliable solution is to engineer the system so that override is operationally costly: requiring multi-step confirmation, documentation of the rationale, and ideally a delay between the override decision and execution. The friction reduces the rate of behavioural override without entirely eliminating the option for legitimate cases.

5.2 *The disclosure implication*

A second implication concerns the relationship between marketed backtests and live performance for systems sold to retail users. A backtest assumes mechanical execution; live performance with override permission systematically underperforms the backtest by 200–400bp. Vendors selling systems should either: (a) disclose this gap explicitly in their marketing materials, or (b) deliver the system with mechanical execution mandated (typically through API integration that bypasses manual intervention).

The current retail systematic-strategy market does not, in general, follow either practice. Backtests are presented as if they will be replicated in live trading; the systems are delivered with full manual override permission; the realised gap between backtested and live performance is typically attributed to "market regime change" rather than to the predictable behavioural mechanism documented in the literature. Honest practice requires bridging this gap.

5.3 *The "vendor runs it" implication*

A stricter implication, sometimes encountered in institutional asset management but rare in retail, is that the only credible disclosure of a system's live performance is the system as run by the vendor on real capital. Backtests can be data-snooped; sample-period live performance can be cherry-picked; signals delivered to clients can be implementation-shortfall-degraded. The only fully reliable evidence that a system produces in live trading what its backtest produces in historical simulation is for the vendor to be running the same system, with the same code, on the same broker, with real capital, in parallel with the client's implementation.

This standard is restrictive but defensible. It eliminates the override-cost problem mechanically (the vendor's own implementation provides the mechanical-execution benchmark), it eliminates the disclosure asymmetry (the vendor's results are observable), and it aligns vendor and client incentives (the vendor's losses on their own capital are real and documentable). The cost to the vendor is non-trivial — running real capital in parallel with client portfolios introduces capacity, regulatory, and operational complexity — but the credibility benefit is substantial.

06. Conclusion

The empirical record on discretionary override of systematic trading rules is consistent across populations, time periods, and study designs: override costs 200–400bp of annualised performance, with the gap widening during

stress periods. The behavioural mechanisms are well-documented (overconfidence, recency, loss aversion, confirmation bias) and reliably present across trader populations. The conditions under which override adds value are narrow and rarely realised in practice.

The system-design implications follow directly. Mechanical execution — the elimination of override as an operational option — is the only reliable mechanism for closing the gap between backtested and live performance. Asking the trader to be disciplined enough to refrain from override in the moments when override is most tempting is asking for behavioural performance the literature suggests is rare. Engineering the system so override is operationally costly or unavailable produces more reliable outcomes than relying on trader discipline.

Limitations. The studies we cite use varying definitions of "override" and varying methodologies for measuring its cost. The 200–400bp range is therefore a synthesis estimate rather than a precise empirical magnitude. Override-cost studies in non-Western markets are sparse; we have not addressed whether the behavioural mechanisms operate identically across cultures. Most importantly, our analysis treats override as a binary decision (taken or not taken); in practice, override often involves partial position-size adjustments rather than complete signal rejection, which we do not model separately.

The principle that follows. A trading system whose performance depends on the trader's discipline to refrain from override is not a system; it is a discipline test that happens to use trading rules as its substrate. A system that produces its documented performance reliably is one whose execution is mechanical — same code, same broker, same risk management — without permitting the trader to introduce their behavioural biases between signal and execution. The math does not care about the trader's feelings about the trade; the system that delivers the math's results does not give the feelings an opportunity to express themselves operationally.

*Override costs **200–400bp per year**. The behavioural mechanisms that drive it are reliably present in every trader population studied. The reliable solution is not better discipline; it is removing the option.*

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APX. Appendix A: Cross-study comparison

TABLE A1. OVERRIDE COST ESTIMATES ACROSS STUDIES

Study	Population	Period	Estimated cost (bp/yr)
Faith (2007), Turtle program	15 trained systematic traders	1983–1987	400
Fung-Hsieh (2001)	Trend-following CTA universe	1989–1997	250
Coval-Shumway (2005)	Prop-desk traders, major U.S. bank	1998–2002	220
Bhansali-Davis (2010)	CTA, normal vs stress	1994–2008	100 / 700
Barber-Odean (2000)	Retail brokerage accounts	1991–1996	200 (residual)

The cost estimates cluster in the 200–400bp range across studies, with the Bhansali-Davis stress-period estimate of 700bp providing the upper bound for high-stress conditions. The cross-study consistency is unusual for a behavioural finance topic, suggesting that the underlying mechanism is robust across populations and conditions.

A.2 Override frequency by trader type

A.4 ProRealTime: mechanical-vs-override comparison stub

The implementation logic discussed above, expressed in ProRealTime's ProBuilder/ProBacktest syntax. The code is illustrative — production deployments include additional risk-management and order-handling logic not shown here.

```
// Run two parallel ProBacktest instances on the same instrument:
// - Instance 1: pure mechanical signal, no override
// - Instance 2: same signal but with a simulated discretionary filter
// Compare equity curves; difference = override cost

DEFPARAM CumulateOrders = False
DEFPARAM PreLoadBars = 270

// Mechanical signal (Instance 1)
mom = (close - close[252]) / close[252]
mechSignal = mom > 0

// Simulated "override" filter (Instance 2):
// Skip trades during recent drawdown, mimicking common override pattern
peak = highest[60](close)
recentDD = (close - peak) / peak
overrideActive = recentDD < -0.05

discSignal = mechSignal AND NOT overrideActive

// In the override version, you skip the entry if discSignal is False
// Even though mechSignal would have entered. This is the modelled override.

IF discSignal AND NOT LongOnMarket THEN
    BUY 1 CONTRACT AT MARKET
ENDIF
IF NOT mechSignal AND LongOnMarket THEN
    SELL AT MARKET
ENDIF
```

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TABLE A2. ESTIMATED OVERRIDE FREQUENCY BY TRADER POPULATION

Population	Override frequency
Disciplined institutional CTA	3–5%
Average institutional CTA	10–15%
Retail systematic-strategy user	25–35%
Retail trader claiming systematic approach	50%+

The progression in override frequency from institutional to retail tracks the progression in observed underperformance. Retail traders who self-describe as "systematic" but actually override more than half their signals are, in operation, discretionary traders with a systematic prior — and produce realised performance characteristic of discretionary trading rather than systematic trading.

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PUBLISHED BY

ProRealAlgos Research
prorealalgos.com

AUTHOR

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Founder, ProRealAlgos

REVISION HISTORY

v1.0	MAY 2021	Initial publication.
v1.1	AUG 2022	Thanks to Curtis for the pointer to the Coval-Shumway paper, now in Section 2.3.
v1.2	MAR 2024	Added the appendix tables on override frequency by trader type.

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